

SVM

11 May 2023 15:02

1. What type of machine learning task is resolved using SVM?

→ Supervised

Consider the $n \times d$ data matrix

$$D = \begin{pmatrix} \mathbf{x}_1 & X_{11} & X_{12} & \dots & X_{1d} \\ \mathbf{x}_2 & X_{21} & X_{22} & \dots & X_{2d} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \mathbf{x}_n & X_{n1} & X_{n2} & \dots & X_{nd} \end{pmatrix} \begin{matrix} \checkmark \\ \\ \\ \checkmark \end{matrix} \begin{matrix} \\ \\ \\ \text{discrete} \end{matrix}$$

where each point $\mathbf{x}_i \in \mathbb{R}^d$ and each attribute $X_j \in \mathbb{R}^d$.

2. What type of classification task is implemented using SVM?

→ Binary Classification



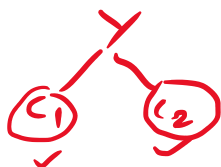
1. What is the decision boundary in SVM and its mathematical/vector representation?

→ Hyperplane

→ 2-D - straight line → $y = mx + c$

→ 3-D - Plane → $ax + by + cz + d = 0$

$$\begin{pmatrix} \underline{a} & \underline{b} & \underline{c} \end{pmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} + d = 0$$



→ $\mathbf{x} \in \mathbb{R}^d$ $h(\mathbf{x}) = 0$

$$h(\mathbf{x}) = \tilde{\mathbf{w}}^T \tilde{\mathbf{x}} + b$$

$$\mathbf{w} = \{w_1, w_2, \dots, w_d\}$$

$$\mathbf{x} = \{x_{11}, x_{12}, \dots, x_{1d}\}$$

$$w_1 x_1 + w_2 x_2 + \dots + w_d x_d + b = 0$$

2. How do you visualize a hyperplane in a 3-dimensional space?

1. Why it is said that the weight vector is normal/orthogonal to the hyperplane, proof?

$$h(\mathbf{x}) = \tilde{\mathbf{w}}^T \tilde{\mathbf{x}} + b = 0$$

Let $\mathbf{n}_1, \mathbf{w}_1$

$$w_1 x_1 + b = 0$$

$$w_1 x_1 = -b$$

$$x_1 = -b/w_1 \quad (w_1 \neq 0)$$

∴ x_1 lies on the hyperplane

$$n_i = -b/w_i \quad (w_i \neq 0)$$

$$\rightarrow (-b/w_1, 0, 0, \dots, 0) \text{ lies on the hyperplane}$$

$$\rightarrow d\text{-dimension}$$

$$\rightarrow (-b/w_1, 0, 0, \dots, 0) \quad (w_1 \neq 0)$$

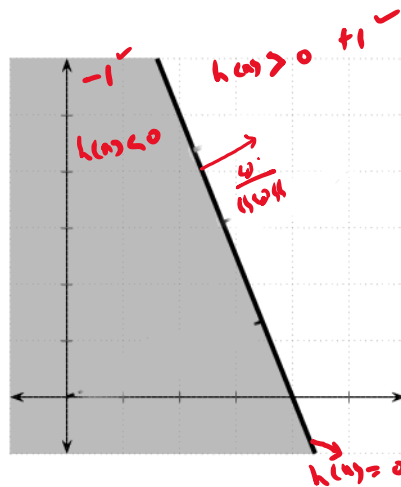
$$\rightarrow \vec{x}_1, \vec{x}_2$$

$$h(\vec{x}_1) = \vec{w}^T \vec{x}_1 + b = 0 \rightarrow \textcircled{1}$$

$$h(\vec{x}_2) = \vec{w}^T \vec{x}_2 + b = 0 \rightarrow \textcircled{2}$$

$$\vec{w}^T (\vec{x}_2 - \vec{x}_1) = 0$$

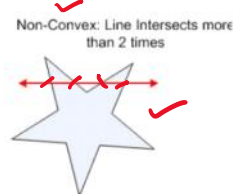
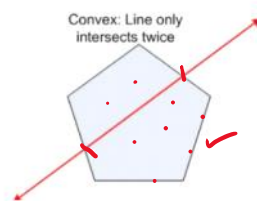
2. What are the classes defined in an SVM based on the hyperplane?



7. What is convex hull?

✓ Convex Polygon: A shape is said to be a convex polygon if a line drawn on that shape intercepts only in 2 points.

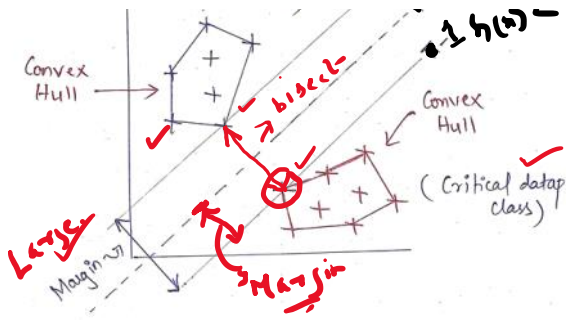
Convex and Non-Convex Polygon



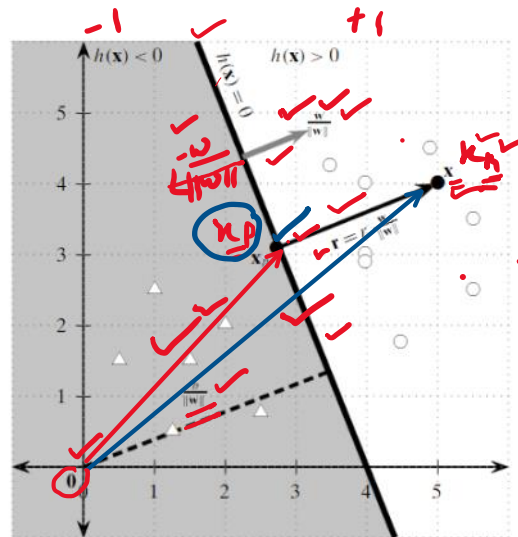
Convex Hull : A smallest shape/polygon where all the datapoints lie inside/on the polygon. ✓

8. Why we call classification task using SVM as large margin Classifier?





9. What is the distance of a point to the hyperplane, how to calculate it?



$$\vec{r} = \vec{x} \cdot \frac{\vec{w}}{\|\vec{w}\|}$$

$\vec{r}$: directed distance
observed

$$x_n = \vec{x}_p + \vec{r}$$

$$\begin{aligned} x_n &= \vec{x}_p + \vec{r} \\ &= \vec{x}_p + r \cdot \frac{\vec{w}}{\|\vec{w}\|} \end{aligned}$$

$$h(x_n) = h\left(\vec{x}_p + r \cdot \frac{\vec{w}}{\|\vec{w}\|}\right)$$

$$h(w) = \vec{w}^T \vec{x} + b$$

$$= \vec{w}^T \left(\vec{x}_p + r \cdot \frac{\vec{w}}{\|\vec{w}\|} \right) + b$$

$$= \vec{w}^T \cdot \vec{x}_p + \vec{w}^T \cdot r \cdot \frac{\vec{w}}{\|\vec{w}\|} + b$$

$$= \vec{w}^T \cdot \vec{x}_p + b + \frac{\vec{w}^T \cdot \vec{w}}{\|\vec{w}\|} \cdot r$$

$$= h(\vec{x}_p) + \frac{\|\vec{w}\|^2}{\|\vec{w}\|} \cdot r = \|\vec{w}\| \cdot r$$

10. What is the distance of the origin to the hyperplane?

=

$$x_0 = \vec{0}$$

$$r = \frac{h(x_n)}{\|\vec{w}\|}$$

$$r = \frac{h(x_0)}{\|\vec{w}\|} = \frac{h(\vec{0})}{\|\vec{w}\|}$$

$$= \frac{\vec{w}^T \cdot \vec{0} + b}{\|\vec{w}\|} = \frac{b}{\|\vec{w}\|}$$

Why the distance calculated in Q9 is called signed distance?

$$h(w) < 0 \rightarrow -w/\|w\|$$

Why the distance calculated in Q9 is called signed distance?

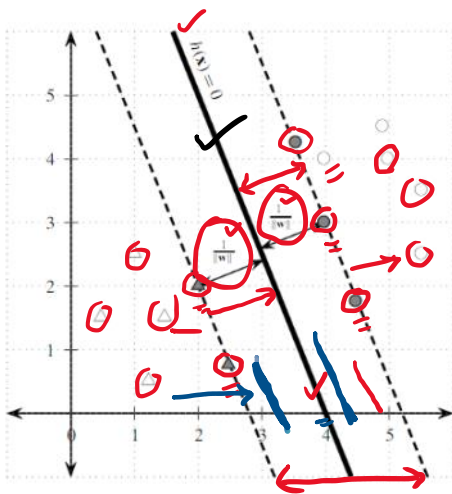
$$\checkmark \quad \tau = \frac{h(x_n)}{\|w\|} \rightarrow \frac{v}{\|w\|} \quad \begin{array}{l} h(x) < 0 \rightarrow \frac{-w}{\|w\|} \\ h(x) > 0 \rightarrow \frac{+w}{\|w\|} \end{array}$$

How to convert the signed distance into unsigned distance?

$$\checkmark \quad \checkmark \quad \checkmark \quad y \rightarrow \text{Label} \begin{matrix} +1 \\ -1 \end{matrix} \quad \checkmark \quad y \cdot \tau = \underbrace{(y)}_{\checkmark} \cdot \underbrace{\frac{h(x_n)}{\|w\|}}_{\checkmark}$$

What is margin and large margin?

1. What is the objective function of large margin classifier, realize it using necessary transformations?



Steps to follow:

1. Setting the lowest margin or the absolute distance as '1' or unit distance for the support vectors from the hyperplane using canonical hyperplanes
2. Setting the constraints or inequalities
3. Choosing a hyperplane that Maximizes the large-margin among all the possible hyperplanes as the objective function.

① $\Rightarrow x_i, y$

$$s^* = \frac{h(x_i) \cdot y}{\|w\|} \rightarrow \begin{array}{l} \text{Absolute Dist} \\ \text{Relative Dist} \end{array}$$

ii) x^* is a support vector.

$$s^* = \frac{h(x^*) \cdot y}{\|w\|} \rightarrow s^* = \frac{1}{\|w\|}$$

iii) $y = 2x + 3$

$$2y = 4x + 6$$

$$\sum h(x^*) \cdot y = 1$$

Canonical hyperplane $S = \frac{1}{h(x^*) \cdot y}$

$$\frac{1}{h(x^*) \cdot y} = \frac{1}{\|w\|}$$

$$\Rightarrow \begin{cases} h(x^*) \cdot y = 1 \\ h(x^*) \cdot y \geq 1 \end{cases} \quad \begin{cases} h(x^*) \cdot y \geq 1 \\ (\omega^T \cdot x^* + b) \cdot y \geq 1 \end{cases}$$

$\Rightarrow$ For all point $x_i \in \mathbb{R}^2$

$$\checkmark \quad \checkmark \quad \checkmark \quad \frac{1}{\|w\|} \quad \checkmark$$

$$\checkmark \quad \alpha_i ((\mathbf{w}^T \mathbf{x}_i + b) \cdot y_i) - 1 = 0$$

$\checkmark \quad \alpha_i \neq 0 \Rightarrow \dots \rightarrow \text{s.v.}$
 $\underline{\underline{\alpha_i = 0}} \Rightarrow \dots \rightarrow \text{N.s.v.}$

$$\rightarrow ((\mathbf{w}^T \mathbf{x}_i + b) \cdot y_i) - 1 = 0 \rightarrow \text{s.v.}$$

$$\rightarrow \alpha_i \geq 0 \quad \checkmark$$

1. What do you mean by dual, how do you realize it in SVM?

$$= \frac{1}{2} \|\mathbf{w}\|^2 - \sum_{i=1}^n \alpha_i \mathbf{w}^T \mathbf{x}_i \cdot y_i - \sum_{i=1}^n \alpha_i \cdot b \cdot y_i + \sum_{i=1}^n \alpha_i$$

$$= \frac{1}{2} \mathbf{w}^T \cdot \mathbf{w} - \mathbf{w}^T \sum_{i=1}^n \alpha_i y_i \mathbf{x}_i - b \sum_{i=1}^n \alpha_i y_i + \sum_{i=1}^n \alpha_i$$

$$\Rightarrow \frac{1}{2} \mathbf{w}^T \cdot \mathbf{w} - \mathbf{w}^T \cdot \mathbf{w} = 0 + \sum_{i=1}^n \alpha_i$$

$$= \sum_{i=1}^n \alpha_i - \frac{1}{2} \|\mathbf{w}\|^2 \quad \checkmark$$

$$\checkmark \quad \checkmark \quad L_{\text{dual}} = \sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j y_i y_j (\mathbf{x}_i \cdot \mathbf{x}_j)$$

$\alpha_i \geq 0, \sum_{i=1}^n \alpha_i y_i = 0$
 $\underline{\underline{= b}}$

Max $\alpha_i \geq 0$ Min $\frac{\|\mathbf{w}\|^2}{2}$ w.b.

$\rightarrow (\mathbf{w}^T \mathbf{x}_i + b) \cdot y_i \geq 1$

19. What is final result of SVM classifier for a new point?

Once we know α_i for all points, we can compute the weight vector $\mathbf{w}$ using Eq. (21.11), but by taking the summation only for the support vectors:

$$\mathbf{w} = \sum_{\alpha_i > 0} \alpha_i y_i \mathbf{x}_i \quad \checkmark \quad (21.14)$$

In other words, $\mathbf{w}$ is obtained as a linear combination of the support vectors, with the $\alpha_i y_i$'s representing the weights. The rest of the points (with $\alpha_i = 0$) are not support vectors and thus do not play a role in determining $\mathbf{w}$.

To compute the bias b , we first compute one solution, denoted b_i , per support vector, as follows:

$$\alpha_i (y_i (\mathbf{w}^T \mathbf{x}_i + b_i) - 1) = 0$$

$$y_i (\mathbf{w}^T \mathbf{x}_i + b_i) = 1$$

$$b_i = \frac{1}{y_i} - \mathbf{w}^T \mathbf{x}_i = y_i - \mathbf{w}^T \mathbf{x}_i \quad \checkmark \quad (21.15)$$

We can then compute b as the average bias value over all the support vectors:

$$b = \text{avg}_{\alpha_i > 0} \{b_i\} \quad \checkmark \quad (21.16)$$

We can then compute b as the average bias value over all the support vectors:

$$b = \text{avg}_{y_i = -1} \{h_i\} \quad (21.16)$$

SVM Classifier

Given the optimal hyperplane function $h(\mathbf{x}) = \mathbf{w}^T \mathbf{x} + b$, for any new point $\mathbf{z}$, we predict its class as

$$\hat{y} = \text{sign}(h(\mathbf{z})) = \text{sign}(\mathbf{w}^T \mathbf{z} + b) \quad (21.17)$$

where the $\text{sign}(\cdot)$ function returns $+1$ if its argument is positive, and -1 if its argument is negative.